BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 038 · Programming Techniques · Sequence and Selection

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does a full adder include that a half adder does not?

Q2. What is a local variable?

Q3. Why does a stack use LIFO?

Q4. Simplify A ∨ (A ∧ B).

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

Recall the term — sequence

Recall the term — selection

Recall the term — Boolean

Recall the term — condition

Recall the term — trace

New learning

Explain the important decision — int score = 7; score = score + 3; if (score >= 10) { Console.WriteLine("Pass"); } outputs Pass.

Worked example

Follow the order of execution

1. Identify the task — Follow the order of execution

2. Follow the logic — int score = 7; score = score + 3; if (score >= 10) { Console.WriteLine("Pass"); } outputs Pass.

3. Check the outcome — Explain how each step contributes to the result. Test an edge case.

We Do · Practise with guidance

Problem 1

value = 3; value *= 4; is value > 10?

Discuss, then respond

Check your thinking

Yes: 12 > 10.

Problem 2

A user enters age 16. Test age >= 18.

False; else branch executes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Apply independently
Trace x = 4; x = x + 7; output x. [2]

Question 2

Apply independently
Write a C# condition that prints Adult when age >= 18. [3]

Question 3

Apply independently
Explain one difference between sequence and selection. [2]

Exit Ticket · Answer before checking

True or false: an else branch runs when its if condition is false.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Carry-in

Q2. A variable declared within a scope

Q3. The most recently inserted item is removed first

Q4. A, by absorption

You Do pages · Independent questions

Q1. 11.

Q2. if (age >= 18) { Console.WriteLine("Adult"); }

Q3. Sequence executes ordered instructions; selection chooses a path based on a condition.

Exit Ticket · Model answer

True. An else branch runs when the related if condition evaluates to false.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: